



NVIDIA Vera Rubin NVL72

Building the next frontier of AI.

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Overview

Seven New Chips, One AI Supercomputer

NVIDIA Vera Rubin NVL72 unifies leading-edge technologies from NVIDIA—72 Rubin GPUs, 36 Vera CPUs, ConnectX®-9 SuperNIC™s, and BlueField®-4 DPUs. It scales up intelligence in a rack-scale platform with the NVIDIA NVLink™ 6 switch and scales out with NVIDIA Quantum-X800 InfiniBand and Spectrum-X™ Ethernet to power the AI industrial revolution at scale. When deployed with NVIDIA Groq 3 LPX racks, Vera Rubin NVL72 delivers a new class of inference performance for trillion-parameter models and million-token context.

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partners, the rack-scale AI supercomputer delivers world-class performance with rapid deployment.

NVIDIA Kicks Off the Next Generation of AI With Rubin

The leading-edge platform scales mainstream adoption, slashing cost per token with five breakthroughs for reasoning and agentic AI models.

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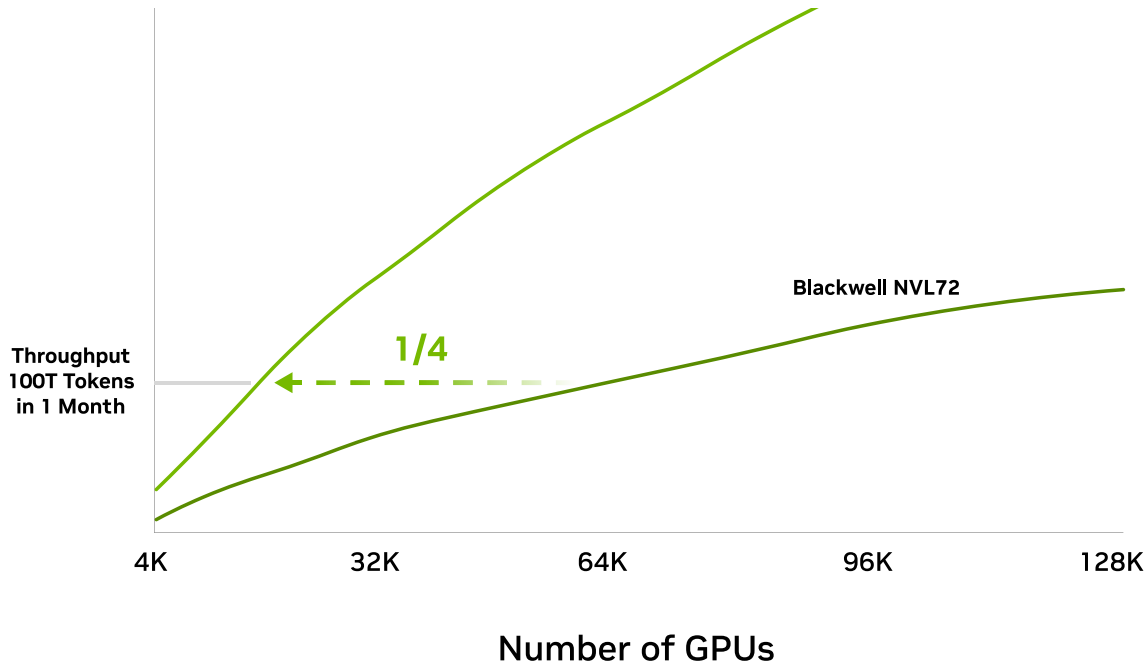
NVIDIA Vera Rubin Opens the Agentic AI Frontier

The NVIDIA Vera Rubin platform offers seven new chips, now in full production, to scale the world's largest AI factories.

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Performance

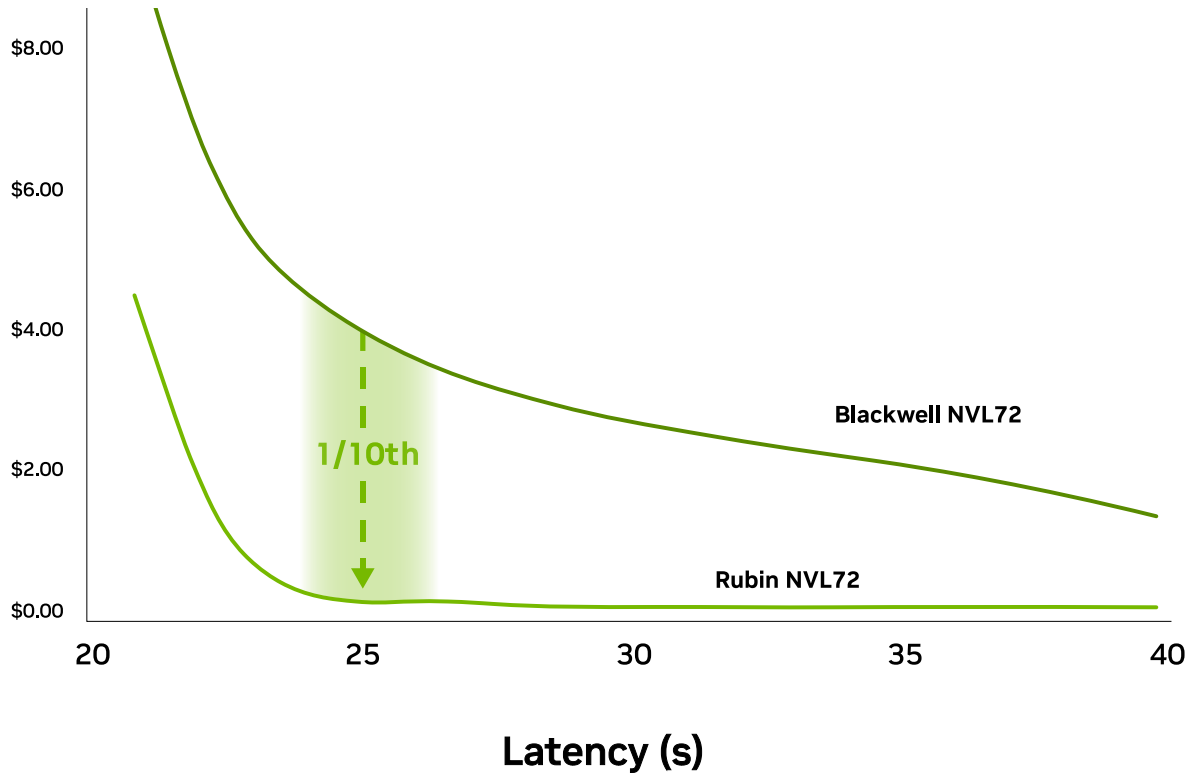
Massive Efficiency Gains in AI Training and Inference



Projected performance subject to change. Number of GPUs based on a 10T MoE model trained on 100T tokens in a fixed timeframe of 1 month.

Boosting Training Efficiency

NVIDIA Rubin trains mixture-of-expert (MoE) models with one-fourth the number of GPUs over the NVIDIA Blackwell architecture.



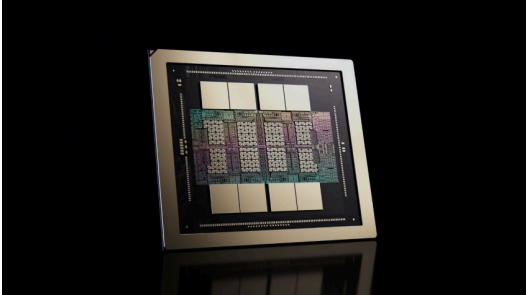
LLM inference performance subject to change. Cost per 1 million tokens based on Kimi-K2-Thinking model using 32K/8K ISL/OSL comparing Blackwell NVL72 and Rubin NVL72.

Driving Down Inference Costs

NVIDIA Rubin delivers one-tenth the cost per million tokens compared to NVIDIA Blackwell for highly interactive, deep reasoning agentic AI.

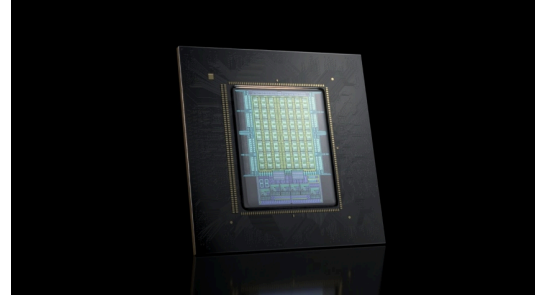
Technology Breakthroughs

Inside the AI Supercomputer

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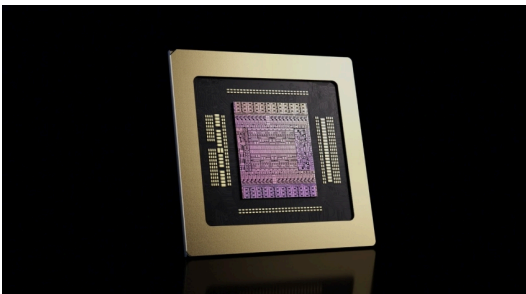
NVIDIA Rubin GPU

Rubin GPUs with HBM4 and 50 PF NVFP4 Transformer Engine made for the next generation of AI.

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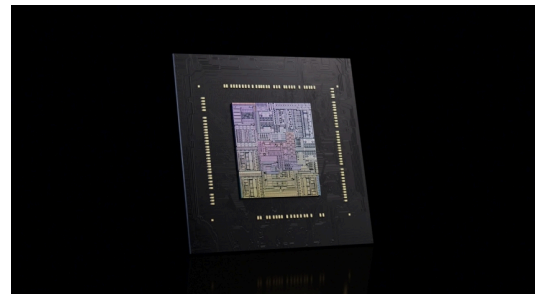
NVIDIA Vera CPU

Vera CPUs are purpose-built for data movement and agentic reasoning, delivering high-bandwidth, energy-efficient compute with deterministic performance.

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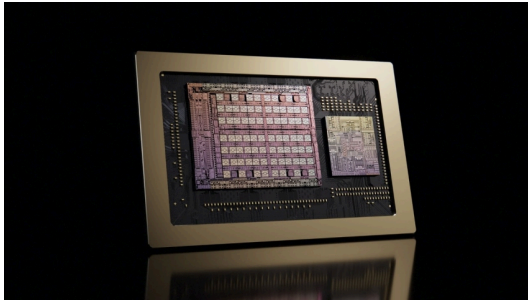
NVIDIA NVLink 6 Switch

NVLink 6 switches feature 3.6 terabytes per second (TB/s) of all-to-all, scale-up bandwidth per GPU,



NVIDIA ConnectX-9 SuperNIC

ConnectX-9 SuperNICs deliver 1.6 terabits per second (Tb/s) of per-GPU bandwidth, with programmable

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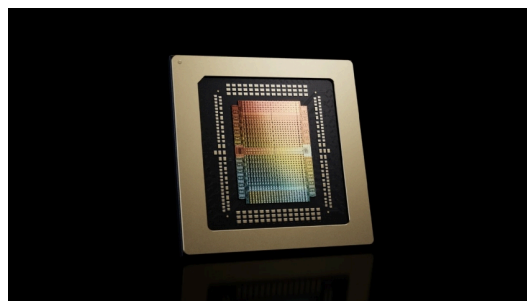
NVIDIA BlueField-4 DPU

BlueField-4 DPUs accelerate data processing across storage, networking, cybersecurity, and elastic scaling in AI factories.

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NVIDIA Spectrum-X Ethernet Co-Packaged Optics

Spectrum-X Ethernet scale-out switches with integrated silicon photonics deliver 5x better power efficiency, 10x higher network resiliency, and up to 5x more uptime over traditional networking with pluggable transceivers.

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NVIDIA Groq 3 LPU

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SRAM, 40 PB/s memory bandwidth, and 640 TB/s scale-up bandwidth per rack. It is co-designed with Vera Rubin NVL72 to deliver 35x inference performance per watt and up to 10x more revenue opportunity for trillion parameter models relative to Blackwell.

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Specifications¹

NVIDIA Vera Rubin NVL72 Specs

	NVIDIA Vera Rubin NVL72	NVIDIA Vera Rubin Superchip	NVIDIA Rubin GPU
Configuration	72 NVIDIA Rubin GPUs 36 NVIDIA Vera CPUs	2 NVIDIA Rubin GPUs 1 NVIDIA Vera CPU	1 NVIDIA Rubin GPU
NVFP4 Inference	3,600 PFLOPS	100 PFLOPS	50 PFLOPS
NVFP4 Training ²	2,520 PFLOPS	70 PFLOPS	35 PFLOPS

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Training			
INT8 ²	18 POPS	0.5 POPS	0.25 POPS
FP16/BF16 ²	288 PFLOPS	8 PFLOPS	4 PFLOPS
TF32 ²	144 PFLOPS	4 PFLOPS	2 PFLOPS
FP32	9,360 TFLOPS	260 TFLOPS	130 TFLOPS
FP64	2,400 TFLOPS	67 TFLOPS	33 TFLOPS
FP32 SGEMM ³	28,800 TFLOPS	800 TFLOPS	400 TFLOPS
FP64 DGEMM ³	14,400 TFLOPS	400 TFLOPS	200 TFLOPS
GPU Memory Bandwidth	20.7 TB HBM4 1,580 TB/s	576 GB HBM4 44 TB/s	288 GB HBM4 22 TB/s
NVLink Bandwidth	260 TB/s	7.2 TB/s	3.6 TB/s
NVLink-C2C Bandwidth	65 TB/s	1.8 TB/s	-
CPU Core Count	3,168 custom NVIDIA Olympus cores (Arm® compatible)	88 custom NVIDIA Olympus cores (Arm compatible)	-

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Total NVIDIA + HBM4 Chips	1,296	30	12
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1. Preliminary information. All values are up to and subject to change.
2. Dense specification.
3. Peak performance using Tensor Core-based emulation algorithms.

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Multi-Instance GPU

NVLink-C2C

NVLink/NVSwitch

Tensor Cores

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